MDS Capstone Proposal - MDP Exception

Student: Om Mehulbhai Patel
UM ID: 69089005
Program: MS in Data Science, University of Michigan
Supervisor: Prof. Ivo Dinov
Course: ENGR 555 (WN26 + FA26, 2+2 credits)

Project Title

Graph Neural Network Framework for Brain Tumor Segmentation and Molecular Subtype Prediction from Multi-Modal MRI

Project Overview

This project develops a novel graph neural network pipeline for simultaneous brain tumor segmentation and molecular subtype prediction using the TCGA-LGG dataset of 110 lower-grade glioma patients. The core research question is whether the spatial and relational structure of a tumor's graph representation, built from multi-modal MRI slices connected across 3D space, encodes information about its underlying molecular identity.

Individual Contributions

I am responsible for the design, implementation, and evaluation of the full technical pipeline, which includes:

Data Pipeline

- Loading and preprocessing 3929 MRI slices across 110 patients from the TCGA-LGG dataset
- Separating three MRI modalities per slice (Pre-Contrast T1, FLAIR, Post-Contrast T1) and performing patient-level z-score normalization
- Identifying and tracking 18 patients with incomplete modality data
- Implementing data augmentation (random flip, rotation, brightness shift, Gaussian noise) producing 1000 additional training slices with correct mask alignment

Graph Construction

- Converting each 256×256 MRI slice into a graph with 64 nodes (32×32 patches) and 420 edges via 8-neighbor connectivity
- Engineering 13-dimensional node feature vectors capturing intensity statistics, texture entropy, skewness, kurtosis, and spatial position
- Building patient-level graphs by stacking all slices and connecting corresponding nodes across adjacent slices through inter-slice edges - the core architectural novelty of this work

Phase 1 - Baseline GCN

- Implementing and training a 2-layer GCN for node-level tumor segmentation
- Achieving a test Dice score of 0.47 as the baseline reference

Phase 2 - Multi-Task ResGCN

- Designing and implementing a 3-layer residual GCN with batch normalization to prevent over-smoothing on large patient graphs
- Training a two-phase strategy: segmentation pre-training followed by fine-tuning, with validation-based early stopping
- Achieving a test Dice of 0.54 and validation Dice of 0.61, a meaningful improvement over the Phase 1 baseline

Genomic Subtype Prediction

- Implementing transfer learning for molecular subtype prediction, extracting 32-dimensional patient embeddings from the trained GNN and training a logistic regression classifier with stratified cross-validation
- Following supervisor guidance to simplify to binary classification (low risk vs high risk) as a proof of concept, achieving 57.9% accuracy above the 50% random chance baseline
- Building a per-patient prediction table showing individual classification outcomes

Ablation Study

- Training the same model on patient graphs without inter-slice edges to isolate their specific contribution
- Confirming that inter-slice edges improve test Dice by 0.03 and precision by 0.10, validating the core architectural claim

3D Visualization

- Building an interactive 3D brain tumor visualization using Plotly with marching cubes surface rendering, showing the tumor surface and inter-slice graph edges for communicating the research to clinical collaborators

Fall 2026 Plans

- Expand binary genomic prediction to 4-class classification across all RNASeq clusters
- Run experiments on Great Lakes HPC cluster for faster iteration
- Implement GNNExplainer interpretability heatmaps showing which brain regions drive molecular predictions
- Extend to survival prediction using Cox proportional hazards on patient embeddings
- Produce a technical report documenting the full pipeline and the findings.

Relevance to MS in Data Science

This project directly applies core MDS competencies including deep learning, graph neural networks, medical image analysis, multi-task learning, and rigorous experimental design with proper train/validation/test splits, ablation studies, and statistical reporting. The work sits at the intersection of machine learning and biomedical data science, producing both technical contributions and clinically meaningful findings. The project demonstrates mastery of MDS core competencies as well as practical skills in HPC cluster computing, reproducible research, and communicating technical work to clinical collaborators.

_____

Prof. Barzan Mozafari (Faculty Advisor)